Stopping-Target Foil Optimization

v3 — fractional-hole geometry, Pareto-HV Bayesian Optimization (qLogNEHVI)

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Mu2e — autoresearch / closed-loop BO

What we optimize

Add **extra foils** up/downstream of the pinned **37-foil** stopping-target base (deployed spec: `rOut=75`, `halfThickness=0.053 mm`, `holeRadius=21.5` — fixed).

6-D knob set (**6 upstream + 6 downstream extras**, one shared triple per side):

knob	range
`extra_rOut_up / dn`	50 – 250 mm
`extra_halfThickness_up / dn`	0.05 – 1.0 mm
`extra_f_up / dn` — hole **fraction** `f = rIn/rOut` (`f<1` ⇒ always buildable)	0 – 0.95

Two competing goals: maximize `S/√B` (Run1A CE significance) and **minimize** `calo` (Run1B calo-stop background). We optimize the **trade-off between them directly** — and report the achievable front.

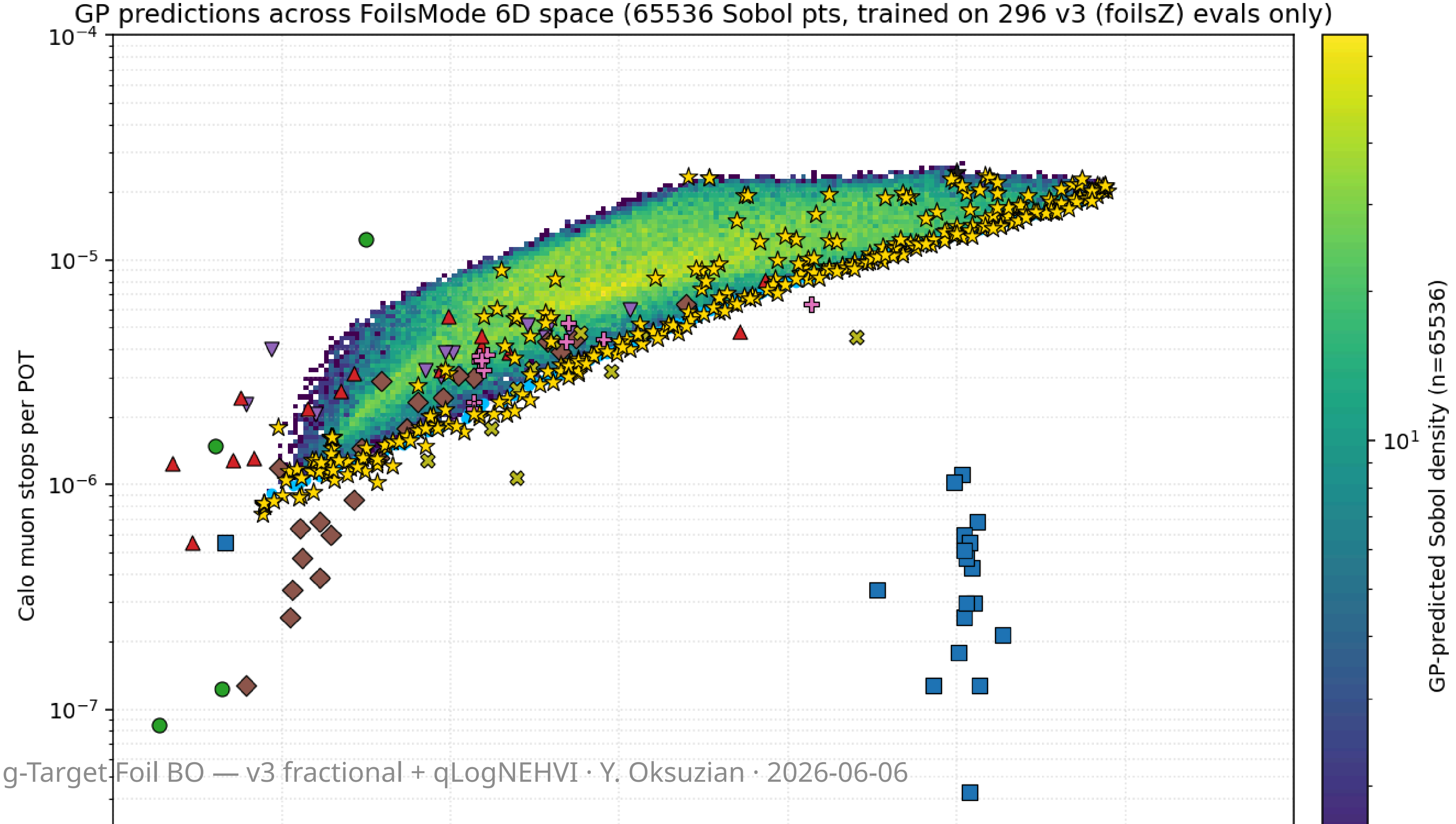
The optimizer: qLogNEHVI

Closed-loop BO: refit a GP on every eval, propose `q` candidates per round, run them in parallel on the grid, repeat.

Acquisition = qLogNEHVI (log Noisy Expected Hypervolume Improvement):

- **Multi-objective** — maximizes the Pareto hypervolume of `(S/√B, −log calo)`; maps the whole `S/√B`–`calo` trade-off front in one run.
- **Noisy** — marginalizes the ~8% calo measurement noise (right for our stochastic G4 metrics).
- **Log-stabilized** — fixes the vanishing-gradient failure of plain qNEHVI, so the candidate optimizer keeps finding good points **even near saturation**.

The 6-D Pareto landscape



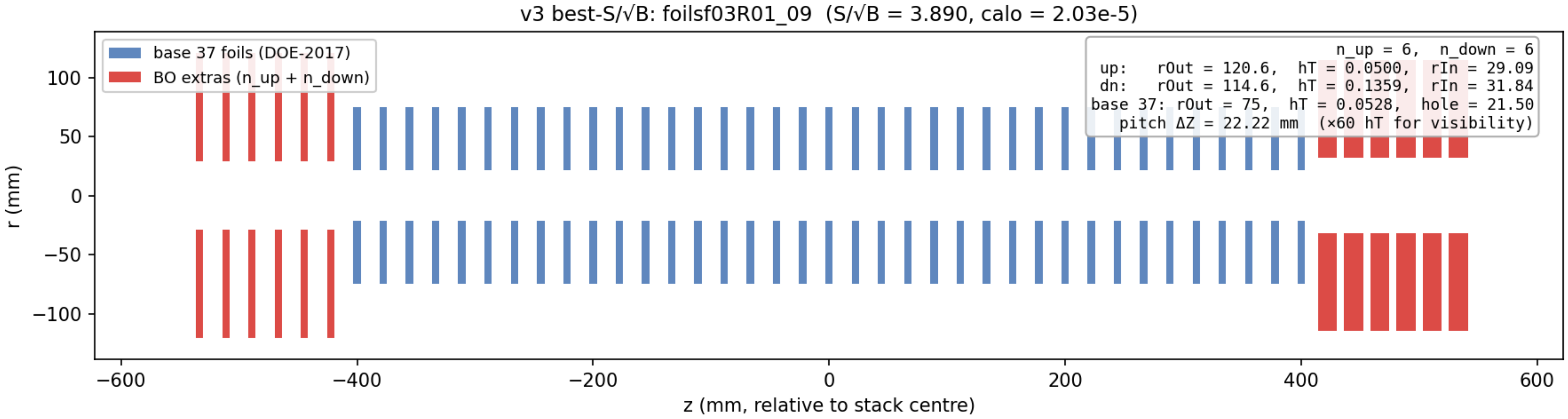
The result: best S/√B at a calo budget

Just *"how much signal if you cap the calo background at B?"* (**297 evals**):

calo budget B	best S/√B	at calo	upstream extras (×6)	downstream extras (×6)
≤ 1e-6 (clean detector)	0.73	8.9e-7	<svg width="48" height="48" viewBox="-35 -35 70 70"><circle cx="0" cy="0" r="30" fill="#3355aa"/><circle cx="0" cy="0" r="9" fill="none" stroke="#cc0000" stroke-width="1.5" stroke-dasharray="2,2"/></svg> solid disc rOut=250 full thick 2.0 mm	<svg width="48" height="48" viewBox="-35 -35 70 70"><circle cx="0" cy="0" r="30" fill="#3355aa"/><circle cx="0" cy="0" r="9" fill="none" stroke="#cc0000" stroke-width="1.5" stroke-dasharray="2,2"/></svg> solid disc rOut=250 full thick 0.99 mm
≤ 1e-5 (**knee**)	**3.01**	9.9e-6	<svg width="48" height="48" viewBox="-35 -35 70 70"><path d="M25.3,0 A25.3,25.3 0 1,0 -25.3,0 A25.3,25.3 0 1,0 25.3,0 M11.6,0 A11.6,11.6 0 1,1 -11.6,0 A11.6,11.6 0 1,1 11.6,0" fill="#3355aa" fill-rule="evenodd"/><circle cx="0" cy="0" r="9" fill="none" stroke="#cc0000" stroke-width="1.5" stroke-dasharray="2,2"/></svg> ring rIn=96.7, rOut=211 thick 0.27 mm	<svg width="48" height="48" viewBox="-35 -35 70 70"><path d="M21.2,0 A21.2,21.2 0 1,0 -21.2,0 A21.2,21.2 0 1,0 21.2,0 M20.1,0 A20.1,20.1 0 1,1 -20.1,0 A20.1,20.1 0 1,1 20.1,0" fill="#3355aa" fill-rule="evenodd"/><circle cx="0" cy="0" r="9" fill="none" stroke="#cc0000" stroke-width="1.5" stroke-dasharray="2,2"/></svg> thin ring rIn=168, rOut=176 thick 0.51 mm
unconstrained (max signal)	**3.89**	2.0e-5	<svg width="48" height="48" viewBox="-35 -35 70 70"><path d="M14.5,0 A14.5,14.5 0 1,0 -14.5,0 A14.5,14.5 0 1,0 14.5,0 M3.5,0 A3.5,3.5 0 1,1 -3.5,0 A3.5,3.5 0 1,1 3.5,0" fill="#3355aa" fill-rule="evenodd"/><circle cx="0" cy="0" r="9" fill="none" stroke="#cc0000" stroke-width="1.5" stroke-dasharray="2,2"/></svg> ring rIn=29.1, rOut=121 full thick 0.10 mm	<svg width="48" height="48" viewBox="-35 -35 70 70"><path d="M13.8,0 A13.8,13.8 0 1,0 -13.8,0 A13.8,13.8 0 1,0 13.8,0 M3.8,0 A3.8,3.8 0 1,1 -3.8,0 A3.8,3.8 0 1,1 3.8,0" fill="#3355aa" fill-rule="evenodd"/><circle cx="0" cy="0" r="9" fill="none" stroke="#cc0000" stroke-width="1.5" stroke-dasharray="2,2"/></svg> ring rIn=31.8, rOut=115 thick 0.27 mm

Sketches: end-on (along beam axis). Filled blue = extra-foil annulus. Dashed red circle = base-foil rOut=75 mm for scale.

Best-significance stack — side view



- **Upstream extras (red, left)**: 6 thin annuli, rIn=29.1 / rOut=120.6, thick 0.10 mm — degrade beam momentum just enough to stop more muons in the base.
- **Downstream extras (red, right)**: 6 thin annuli, rIn=31.8 / rOut=114.6, thick 0.27 mm — pass the unstopped beam through the hole, catch off-axis halo.
- **Base 37 (blue)**: deployed DOE-2017 stack, untouched (rOut=75 / hole=21.5 / thick 0.106 mm).
- Foil thickness ×60 for visibility (real hT << pitch ΔZ = 22.22 mm).

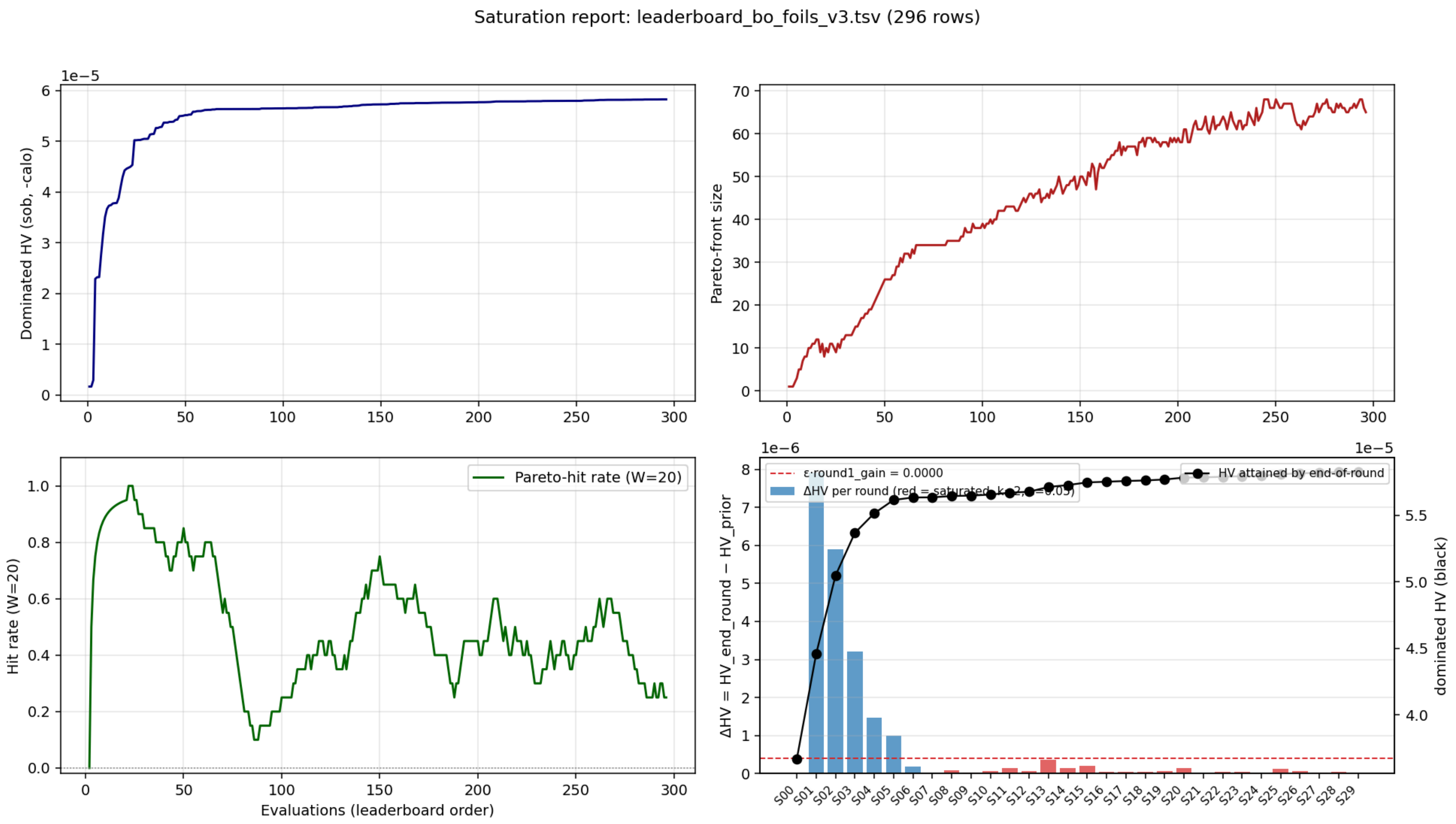
Top 3 by S/√B

rank	config	S/√B	calo	upstream (×6 extras)			downstream (×6 extras)		
				rOut	rIn	hT	rOut	rIn	hT
1	`foilsf03R01_09`	**3.890**	2.03e-5	120.6	29.1	0.050	114.6	31.8	0.136
2	`foilsf07R00_00`	3.880	2.15e-5	111.5	39.0	0.060	114.5	46.7	0.132
3	`foilsf06R04_04`	3.880	2.13e-5	102.0	15.3	0.064	107.3	34.2	0.141

All dimensions mm. Base 37 foils unchanged: rOut=75, rIn=21.5, hT=0.053.

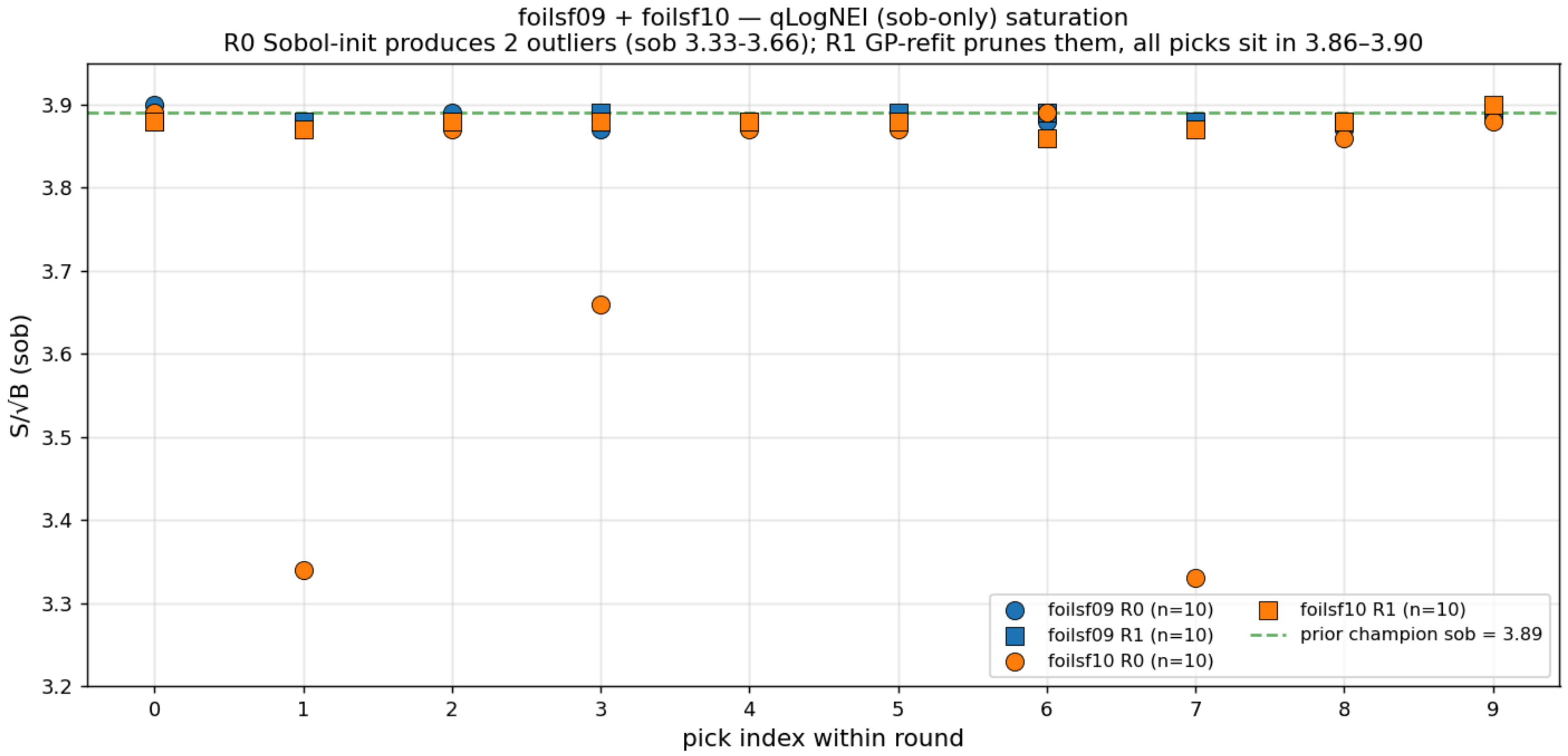
- **All three converged to the same geometry family**: rOut ≈ 115–137 mm, **upstream hT pinned at the lower bound (0.050 mm)** + **downstream slightly thicker** (0.11–0.14 mm), small upstream hole + small downstream hole.
- S/√B values within **0.5%** — the front is **flat at the top**; any of the three is operationally equivalent.
- Picked across **3 separate campaigns** (foilsf03 / foilsf06 / foilsf07), independent restarts — strong evidence the optimum is real, not a single-run artefact.

Convergence — front saturated



VERDICT: SATURATED — picker maps the front rather than climbing it (297 evals, ΔHV near zero for 10+ rounds).

qLogNEI (sob-only) cross-check — foilsf09 + foilsf10



Two independent **single-objective qLogNEI** runs (40 evals, picker drops the DS-off `run1b_mubeam` stage → ~40% faster). R0 Sobol-init produces a few low-rOut outliers (sob 3.33–3.66); R1 GP-refit prunes them — all 20 R1 picks sit in **3.86–3.90**, matching the 8-campaign cross-picker saturation at **sob ≈ 3.89**

Status & next steps

- **Campaign so far:** 297 evals across 15+ rounds of qLogNEHVI (q = 10). GP is under-identified on the small training set (length_scale rails to 1000 mm), but the picker keeps advancing the front.